UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 8B – Job Design I

EIN3241: Human Factors and Ergonomics
Spring 2026

April 15, 2026

Dr. Jad A. Atweh

Admin Issues

- Project Presentations Day 1
  - Friday
  - On all days, teams should have submitted their presentations **BEFORE** class to Canvas.
  - Attendance is mandatory for all presentation dates.
  - Presentation Grade: 9%

Criteria
Timing (2%)
Clarity and coherence (2%)
Addressing questions (2%)
Presentation skills (3%)

Manual Material Handling (MMH)

- Manual material handling (MMH) refers to a variety of techniques related to manually moving objects (e.g. lift, lower, push, pull, carry, etc.).
- MMH is virtually everywhere, and associated injuries are high:
  - 25% of accidents in industry are related to MMH.
  - 60% of overexertion injuries are due to lifting/lowering.
  - 60% of low back pain (LBP) is primarily due to overexertion.
  - LBP related to occupational factors costs $4.6 billion/yr.

Control of Manual Handling Incidents

- Requires a comprehensive program, considering the following factors:

TABLE 3.7 FACTORS AFFECTING MANUAL HANDLING CAPABILITY

Worker characteristics	Material characteristics
Age	Weight
Sex	Bulkiness
Motivation	Load distribution
Physique, etc.	Handles, etc.
Task characteristics	Work organization
Reach requirement	Work–rest cycles
Frequency of handling	Training, selection
Duration, etc.	Job rotation, etc.

Source: Adapted from Ref. [30].

MMH Research Approaches

- Evaluation of injury potential due to MMH follows four major avenues of research:
  1. *Epidemiological approach*: characterizes the job, workplace, and other factors to derive trends leading to injuries; main input: historical records.
  2. *Biomechanical approach*: characterizes forces/moments on the body to estimate task and load characteristics that may lead to injuries. Applicable for tasks where the frequency is less than 4 lifts per min.
  3. *Psychophysical approach*: participants adjust the load/task characteristics to arrive at an “acceptable “ setup.
  4. *Physiological approach*: Evaluates MMH task in terms of cardiovascular stress, e.g. O2 consumption, HR, blood pressure, energy expenditure.

Job Analysis Methods

- Different methods for assessing physical stress in a job that could be harmful to the musculoskeletal system.
  - NIOSH Lifting Equation
  - Biomechanics Software (e.g., 3DSSPP)
  - RULA/RUBA
  - LMD
  - Goniometer
  - Motion Capture Analysis Systems

NIOSH Lifting Equation (1991 Revised)

- The NIOSH lifting equation was developed to help industry combat back injuries that result from lifting tasks.
  - Identifies lifting tasks that may lead to back injuries
  - Shows areas where task can be improved
- Originally developed in 1981 but revised in 1991.
- Equation computes a recommended weight limit (RWL) and compares it to actual weight used.

NIOSH Lifting Equation

- Lifting index (LI) is a ratio of RWL and weight actually lifted (L):

$$LI = \frac{L}{RWL}$$

- Maximum weight of lift under ideal conditions (load constant): 23 kg/ 51 lbs.
- An LI that is:
  - ≤ 1 is good; **action limit (AL)**
  - Between 1 and 3 shows potential problems
  - ≥ 3 Eliminate or revise task immediately; **maximum permissible limit (MPL)**

NIOSH Lifting Equation

- RWL = (LC)(HM)(VM)(DM)(AM)(FM)(CM)

- Where:
  1. LC = load constant
  2. HM = horizontal multiplier
  3. VM = vertical multiplier
  4. DM = distance multiplier
  5. AM = asymmetric multiplier
  6. FM = frequency multiplier
  7. CM = coupling multiplier

This is the Revised NIOSH LE:

- The original 1981 NIOSH equation only used three multipliers (horizontal, vertical, and distance).
- The 1991 Revised Equation expanded it to six multipliers + the load constant.
- All multipliers in the NIOSH equation range from 0 to 1

NIOSH Lifting Equation

		METRIC	U.S. CUSTOMARY
Load Constant	LC	23 kg	51 lb
Horizontal Multiplier	HM	(25/H)	(10/H)
Vertical Multiplier	VM	1-(.003 |V-75|)	1-(.0075 |V-30|)
Distance Multiplier	DM	.82 + (4.5/D)	.82 + (1.8/D)
Asymmetric Multiplier	AM	1-(.0032A)	1-(.0032A)
Frequency Multiplier	FM	From Table 3.15	From Table 3.15
Coupling Multiplier	CM	From Table 3.17	From Table 3.17

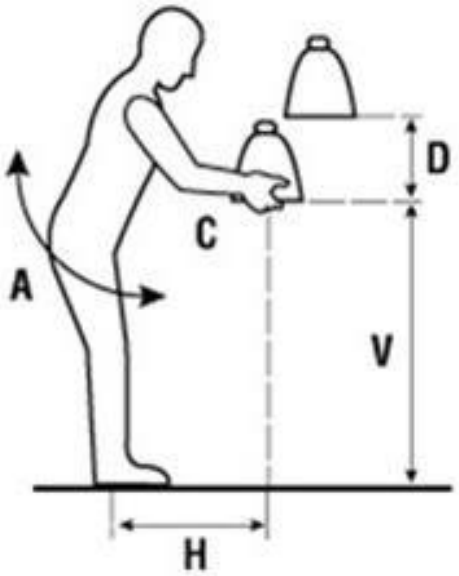
1. Load Constant (LC)

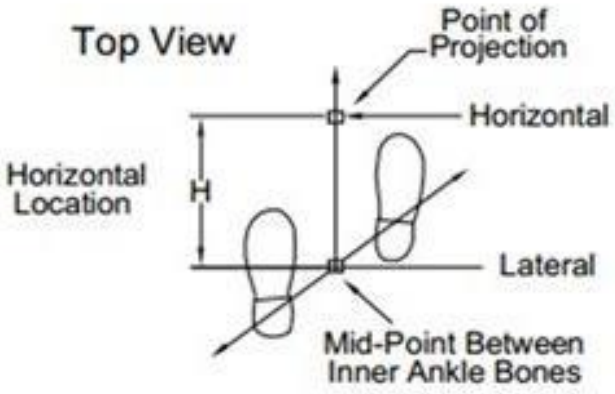
- Load constant (LC) = 23 kg / 51 lbs
- Again, the multipliers in the equation range between 0 and 1.
- Therefore, the RWL can take a maximum value of 23 kg if all multipliers are 1 (ideal lift).

2. Horizontal Multiplier (HM)

- Horizontal (H): distance between center of load and midpoint between ankles.
- $0 \leq HM \leq 1.0$
- If $H < 25$ cm, then $HM = 1$ (ideal)
- If $H > 63$ cm, then $HM = 0$ (unacceptable)

Horizontal Multiplier	(25/H)

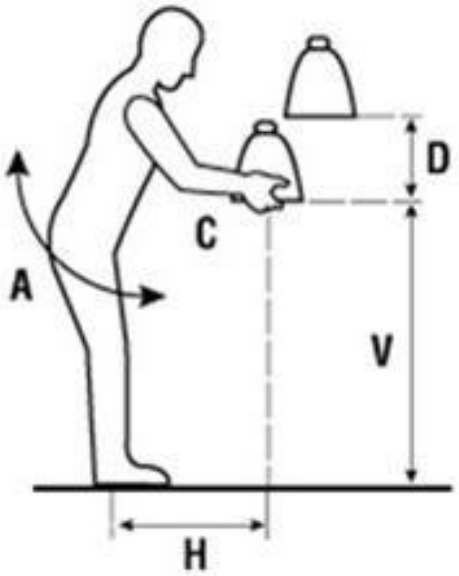




3. Vertical Multiplier (VM)

- Vertical (V): vertical distance of first knuckle of middle finger
- from the floor.
- 0 ≤ VM ≤ 1.0
- If V > 175cm, then VM = 0 (unacceptable)

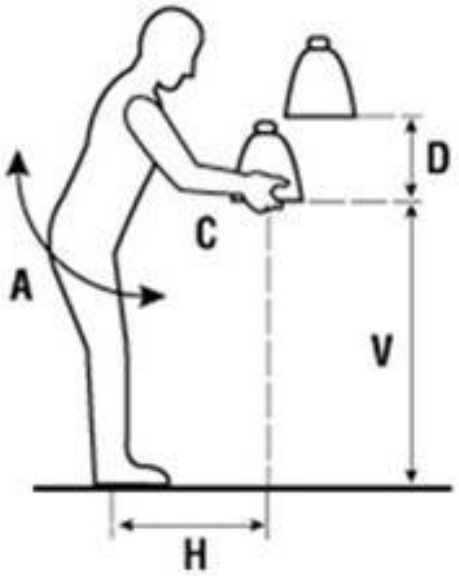
Vertical Multiplier	$1-(.003\,\lvert V-75\rvert)$

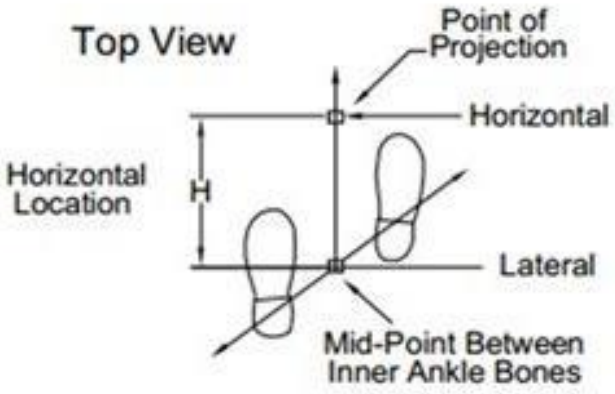


4. Distance Multiplier

- Distance (D): vertical travel distance of the load
- $0 \leq DM \leq 1.0$
- If $D < 25$ cm, then $DM = 1$ (ideal)
- If $D > 175$ cm, then $DM = 0$ (unacceptable)

Distance Multiplier	$.82 + (4.5/D)$

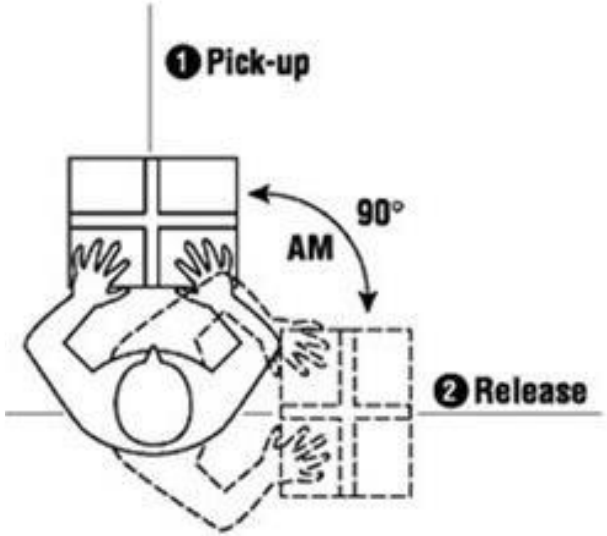




5. Asymmetric Multiplier (AM)

- Asymmetry (A): angle between the location of the load and the mid sagittal plane.
- $0 \leq AM \leq 1.0$
- If $A = 0°$, then $AM = 1$ (ideal)
- If $A > 135°$, then $AM = 0$ (unacceptable)

Asymmetric Multiplier	1-(.0032A)



6. Frequency Multiplier (FM)

- Frequency (F): average number of lifts per minute, as measured over a 15-minute period.
- $0 \leq FM \leq 1.0$
- If $F > 15$ lifts/minute, then $FM = 0$ (unacceptable)
- FM is a function of F, duration, and V (Table 3.15)

6. Frequency Multiplier (FM)

30 inches = $30\ inches \times 2.54 = 76.2\ cm$

A worker lifts a box 6 times per minute, for 2.5 hours, and the vertical height $V = 65\ cm$.

$$FM = 0.27$$

TABLE 3.15 FREQUENCY MULTIPLIER TABLE (*FM*)

Frequency Lifts/min (F)[2]	Work Duration					
	≤1 Hour		>1 but ≤2 Hours		>2 but ≤8 Hours	
	V<30[1]	V≥30	V<30	V≥30	V<30	V≥30
≤0.2	1.00	1.00	.95	.95	.85	.85
.05	.97	.97	.92	.92	.81	.81
1	.94	.94	.88	.88	.75	.75
2	.91	.91	.84	.84	.65	.65
3	.88	.88	.79	.79	.55	.55
4	.84	.84	.72	.72	.45	.45
5	.80	.80	.60	.60	.35	.35
6	.75	.75	.50	.50	.27	.27
7	.70	.70	.42	.42	.22	.22
8	.60	.60	.35	.35	.18	.18
9	.52	.52	.30	.30	.00	.15
10	.45	.45	.26	.26	.00	.13
11	.41	.41	.00	.23	.00	.00
12	.37	.37	.00	.21	.00	.00
13	.00	.34	.00	.00	.00	.00
14	.00	.31	.00	.00	.00	.00
15	.00	.28	.00	.00	.00	.00
>15	.00	.00	.00	.00	.00	.00

[1] Values of *V* are in inches.

[2] For lifting less frequently than once per 5 minutes, set *F* = .2 lifts/minute

7. Coupling Multiplier (CM)

- Coupling (C): is the effectiveness of the hand container (or grip) interface, described as good, fair, or poor.
  - ‘Good’ if object allows hands to wrap around it (i.e. box with handles or hand hold cutouts).
  - ‘Fair’ if object has less than optimal handles or no handles but allows lifter to bend fingers at 90 o during lift.
  - ‘Poor’ if object is difficult to be grasped, with no handles or cutouts.
- $0 \leq CM \leq 1.0$
- CM is a function of C and V (Table 3.16 and 3.17)

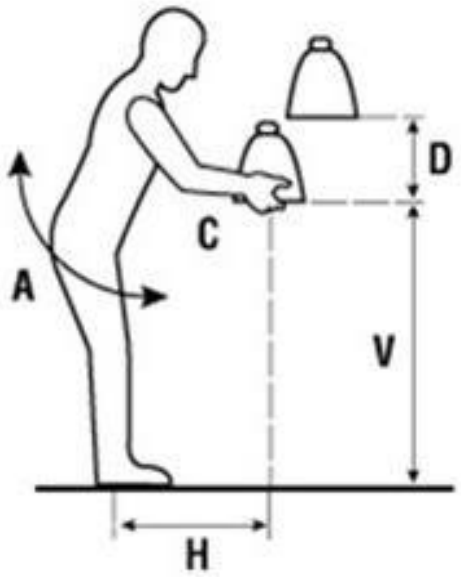
7. Coupling Multiplier (CM)

TABLE 3.17 COUPLING MULTIPLIER

Coupling Type	Coupling Multiplier	
	$V<30$ inches (75 cm)	$V\geq30$ inches (75 cm)
Good	1.00	1.00
Fair	0.95	1.00
Poor	0.90	0.90

NIOSH Lifting Equation

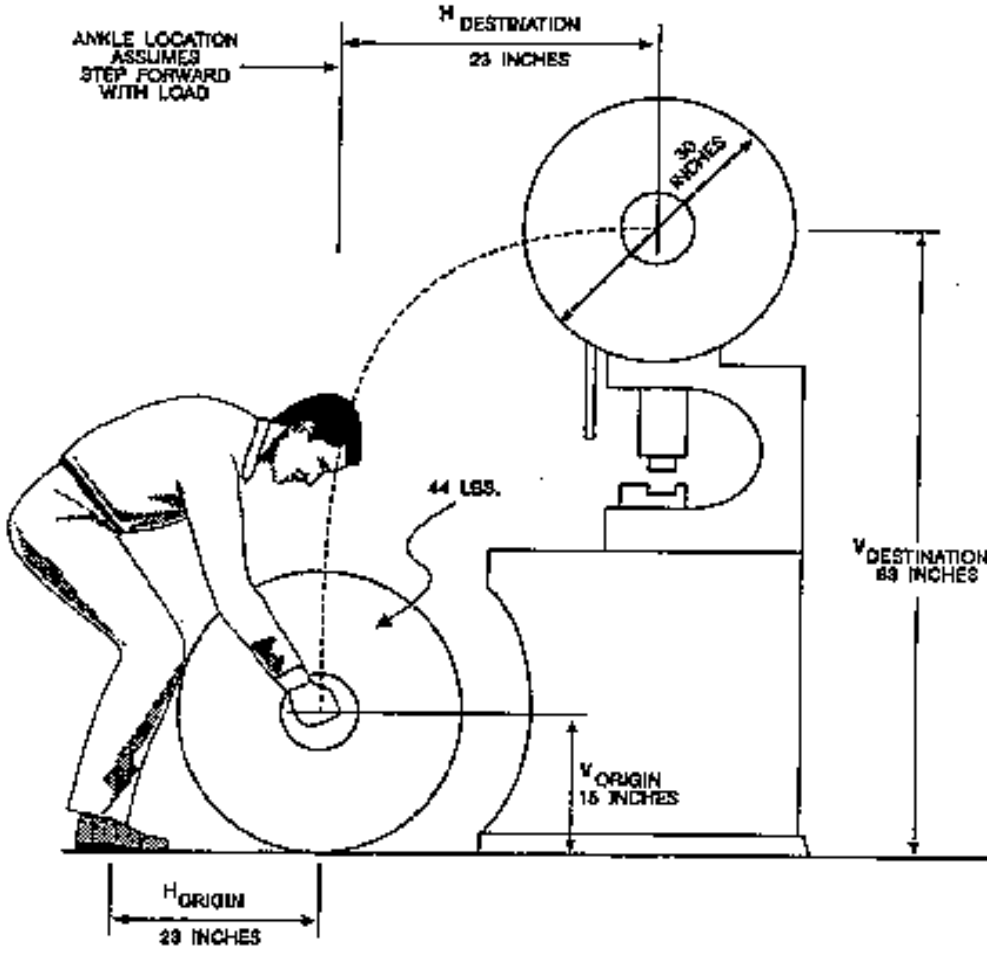
- Must perform the analysis at the origin and destination of the lift.
- Frequency is $\leq 0.2\ lifts/min$ in 30 mins.
- Average load is 10 lbs
- Which RWL will represent the lift?
- What is Lifting Index?



Origin Destination

Example: Loading

- Example: Loading Punch Press Stock (Note measurements are in inches)



STEP 1. Measure and record task variables

Object Weight (lbs)		Hand Location (in)				Vertical Distance (in)	Asymmetric Angle (degrees)		Frequency Rate	Duration	Object Coupling
		Origin		Dest.			Origin	Destination	lifts/min	(HRS)	
L (AVG.)	L (Max)	H	V	H	V	D	A	A	F		C
44	44	23	15	23	63	48	0	0	<.2	<1	Fair

STEP 2. Determine the multipliers and compute the RWL's

RWL = LC × HM × VM × DM × AM × FM × CM

ORIGIN RWL = 51 × .44 × .89 × .86 × 1.0 × 1.0 × .95 = 16.3 Lbs

DESTINATION RWL = 51 × .44 × .75 × .86 × 1.0 × 1.0 × 1.0 = 14.5 Lbs

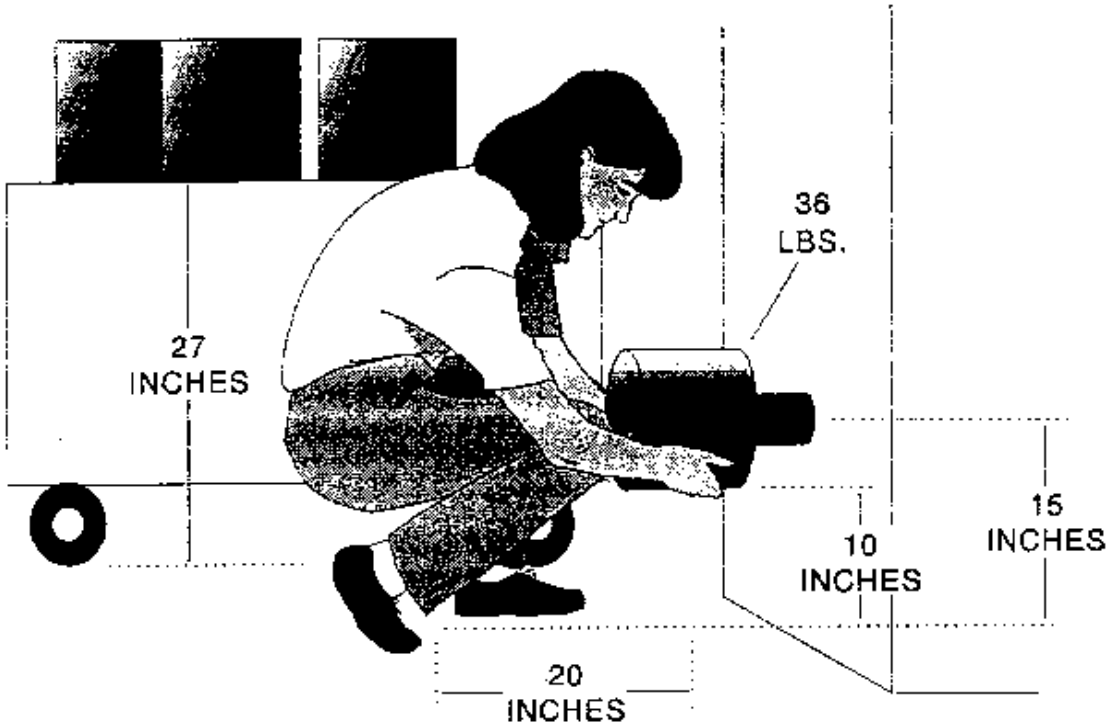
STEP 3. Compute the LIFTING INDEX

ORIGIN LIFTING INDEX = OBJECT WEIGHT (L) / RWL = 44 / 16.3 = 2.7

DESTINATION LIFTING INDEX = OBJECT WEIGHT (L) / RWL = 44 / 14.5 = 3.0

Example: Loading

- Example: Example: Loading Supply Rolls (Note measurements are in inches



STEP 1. Measure and record task variables

Object Weight (lbs)		Hand Location (in)				Vertical Distance (in)	Asymmetric Angle (degrees)		Frequency Rate	Duration	Object Coupling
		Origin		Dest.			Origin	Destination	lifts/min	(HRS)	
L (AVG.)	L (Max.)	H	V	H	V	D	A	A	F		C
35	35	15	27	20	10	17	0	0	<.2	<1	Poor

STEP 2. Determine the multipliers and compute the RWL's

RWL = LC × HM × VM × DM × AM × FM × CM

ORIGIN RWL = 51 × .67 × .98 × .93 × 1.0 × 1.0 × .90 = 28.0 Lbs

DESTINATION RWL = 51 × .50 × .85 × .93 × 1.0 × 1.0 × .90 = 18.1 Lbs

STEP 3. Compute the LIFTING INDEX

ORIGIN LIFTING INDEX = OBJECT WEIGHT (L) / RWL = 35 / 28.0 = 1.3

DESTINATION LIFTING INDEX = OBJECT WEIGHT (L) / RWL = 35 / 18.1 = 1.9

General Design/Redesign Suggestions

If HM is less than 1.0	Bring the load closer to the worker by removing any horizontal barriers or reducing the size of the object. Lifts near the floor should be avoided; if unavoidable, the object should fit easily between the legs.
If VM is less than 1.0	Raise/lower the origin/destination of the lift. Avoid lifting near the floor or above the shoulders.
If DM is less than 1.0	Reduce the vertical distance between the origin and the destination of the lift.
If AM is less than 1.0	Move the origin and destination of the lift closer together to reduce the angle of twist, or move the origin and destination further apart to force the worker to turn the feet and step, rather than twist the body.
If FM is less than 1.0	Reduce the lifting frequency rate, reduce the lifting duration, or provide longer recovery periods (i.e., light work period).
If CM is less than 1.0	Improve the hand-to-object coupling by provinding optimal containers with handles or handhold cutouts, or improve the handholds for irregular objects.

Conditions of Use: NIOSH Lifting Equation

- There are situations where NIOSH lifting equation does NOT apply, such as:
  - One handed-lift
  - Duration > 8 hours
  - Lifting while seated or in restricted workspace
  - Lifting unstable objects
  - Lifting while carrying, pushing, or pulling
  - High speed motion (> 76 cm/s)
  - Unreasonable floor/ foot coupling (< 0.4 coefficient of friction)
  - Unfavorable environment (humidity and temperature)

UF

Herbert Wertheim
College of Engineering
UNIVERSITY *of* FLORIDA

LEADING THE CHARGE, CHARGING AHEAD